

Progress

AI-RAN GPU Isolation — Mechanism Deep-Dive

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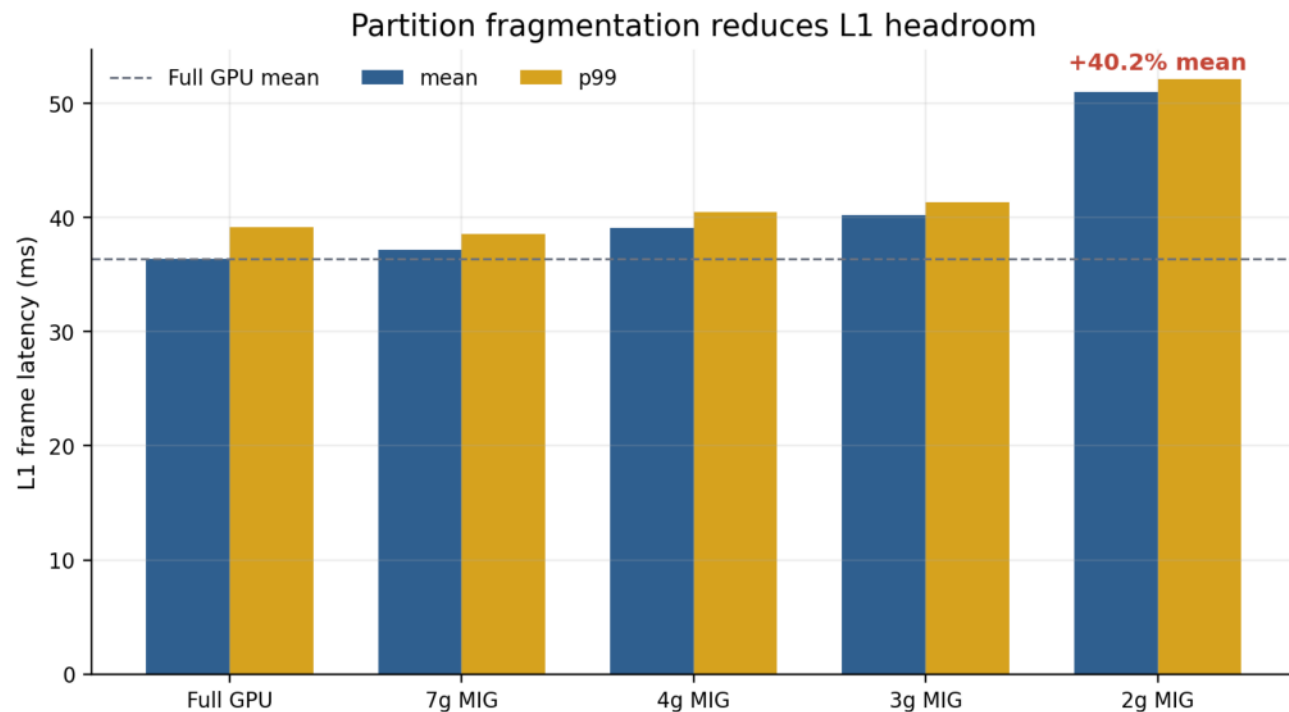
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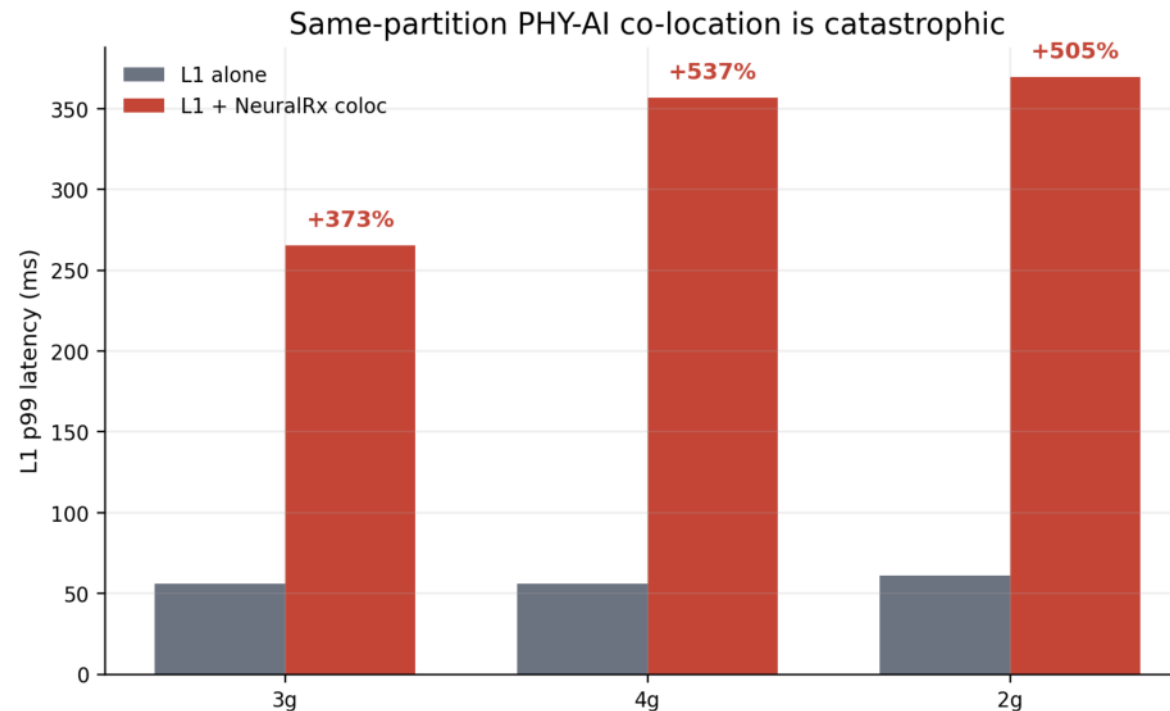
Today

- We extend the previous MIG single-tenant analysis to AI co-tenancy (Exp1-6 on 5/29 covered MIG-alone overhead of +13-32 ms).
- **Question:** Under realistic AI co-location, where does the L1 latency cost originate, and through what hardware mechanism?
- **Setup:** n=500-1000 per condition (statistical power 25× over the prior n=20 sweep), with CloudLab d8545 + Perlmutter cross-validation.



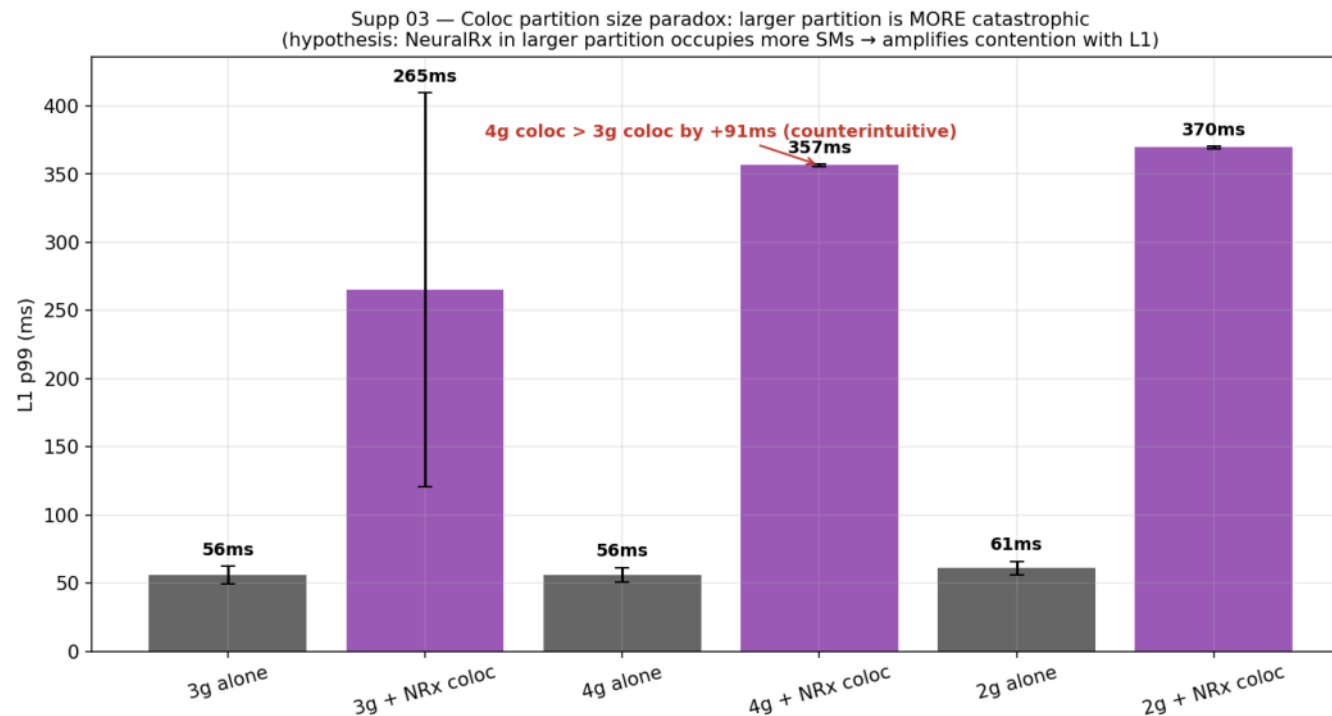
Exp7) Same-Partition Co-location is Catastrophic

- **Catastrophic Collapse:** Co-locating any AI workload in the same MIG partition as L1 inflates p99 latency to ~357 ms — an 8× regression over the cross-partition baseline (~45 ms).
- **Workload-Invariant:** chanpred, NeuralRx, Qwen, xApp, and ResNet all converge to ~357 ms; the failure is not specific to any single AI workload.
- **Placement, Not Workload, Is the Cost Driver:** Reproducible across n=500/condition with std=9 ms — the cost arises from where AI is placed, not from what AI is doing.



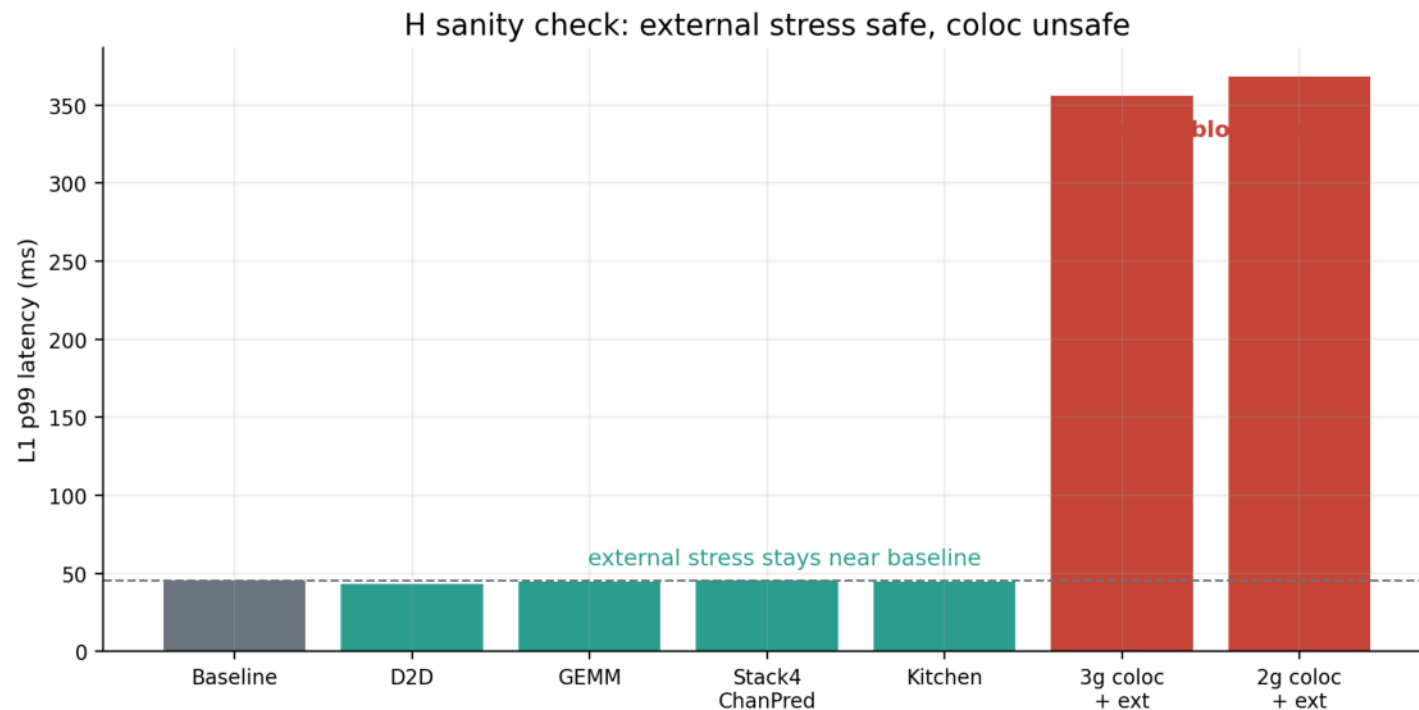
Exp8) Larger Partition = MORE Catastrophic

- **Counter-Intuitive Outcome:** 4g co-location (371 ms) is worse than 2g co-location (369 ms), inverting the assumption that larger slices dilute contention.
- **Resource Allocation Fails as Mitigation:** Adding SMs and HBM to the partition does not reduce collapse; bandwidth share is decisive but not through MIG's partition allocation.
- **Bottleneck Lives Outside the Partition Boundary:** The contended resource is chip-global and unreachable through the per-partition abstraction MIG provides.



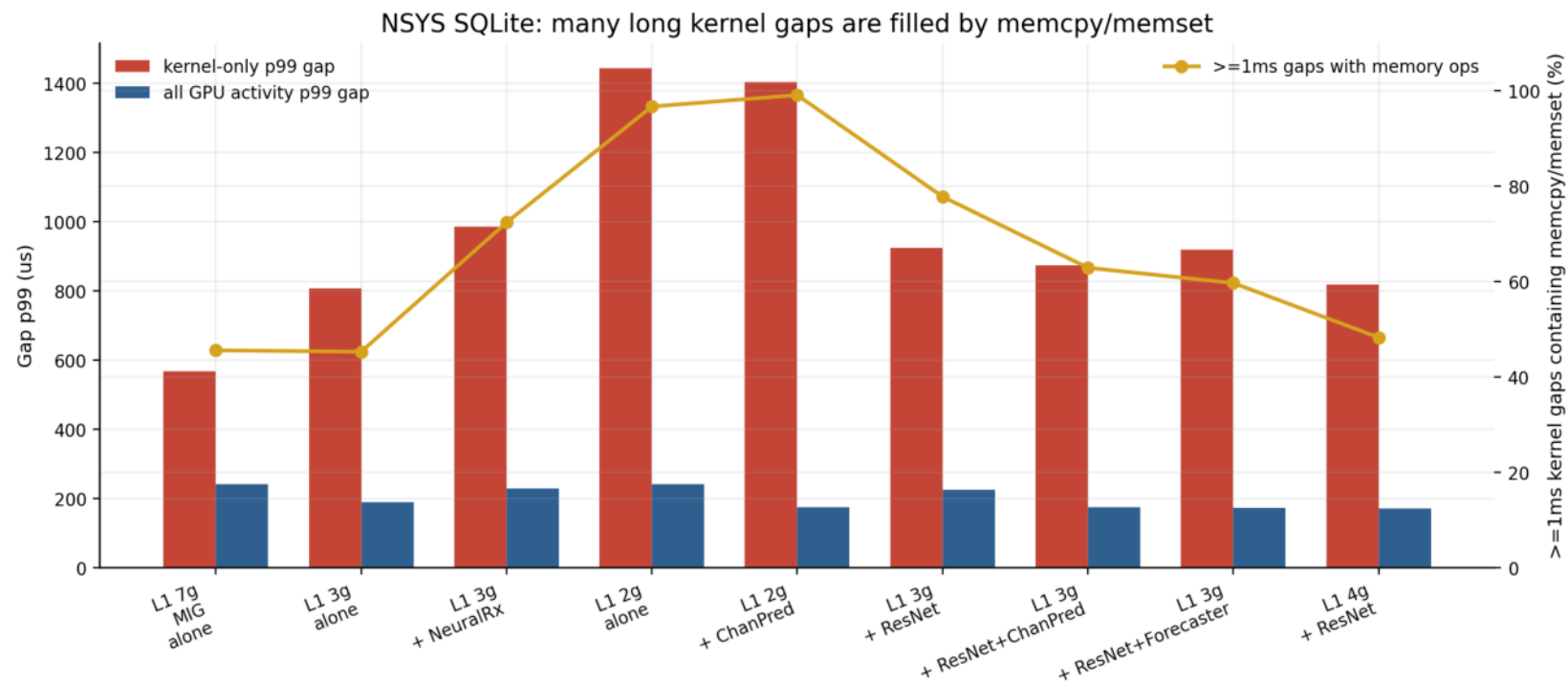
Exp9) Placement is a Binary Decision

- **No Soft Transition:** The same chanpred workload yields 44 ms when cross-partitioned vs 357 ms when co-located — a single configuration switch separates the two regimes.
- **Single Mistake Triggers 8× Penalty:** One placement misconfiguration produces the full collapse; no operational tuning recovers intermediate values.
- **Operational Implication:** The latency cost model is binary rather than gradual; admission control must enforce placement explicitly at deployment time.



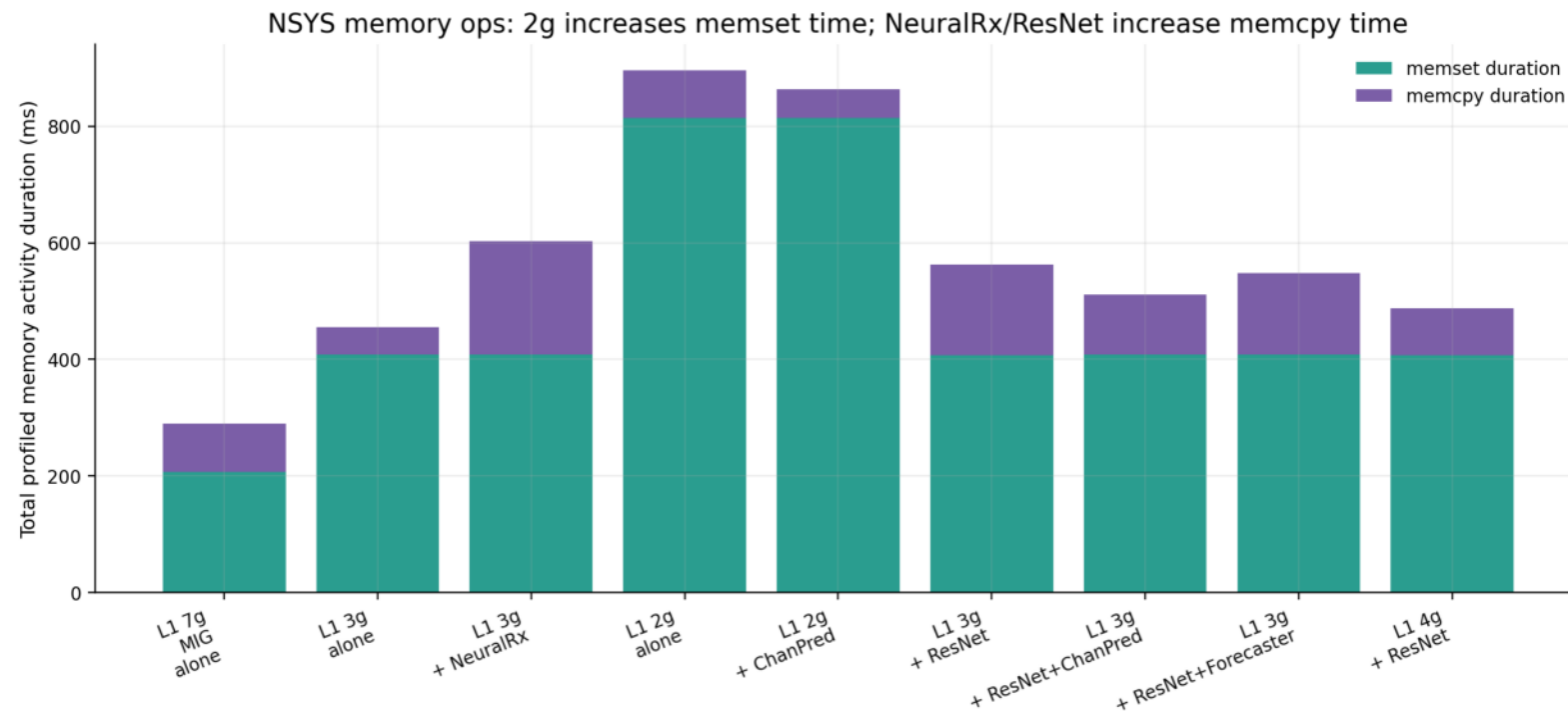
Mechanism Layer 1 — Gap Aligns to Op Boundaries

- **Kernel Duration is Unchanged:** Per-kernel execution time is identical under alone and co-location; only inter-kernel gaps grow.
- **Gaps Are Not Random Idle:** Inter-kernel gaps cluster precisely at memcpy and memset boundaries rather than distributing uniformly across the timeline.
- **Robust Across 30 Conditions:** The boundary localization holds across the full NSYS condition matrix, ruling out a single-capture artifact.



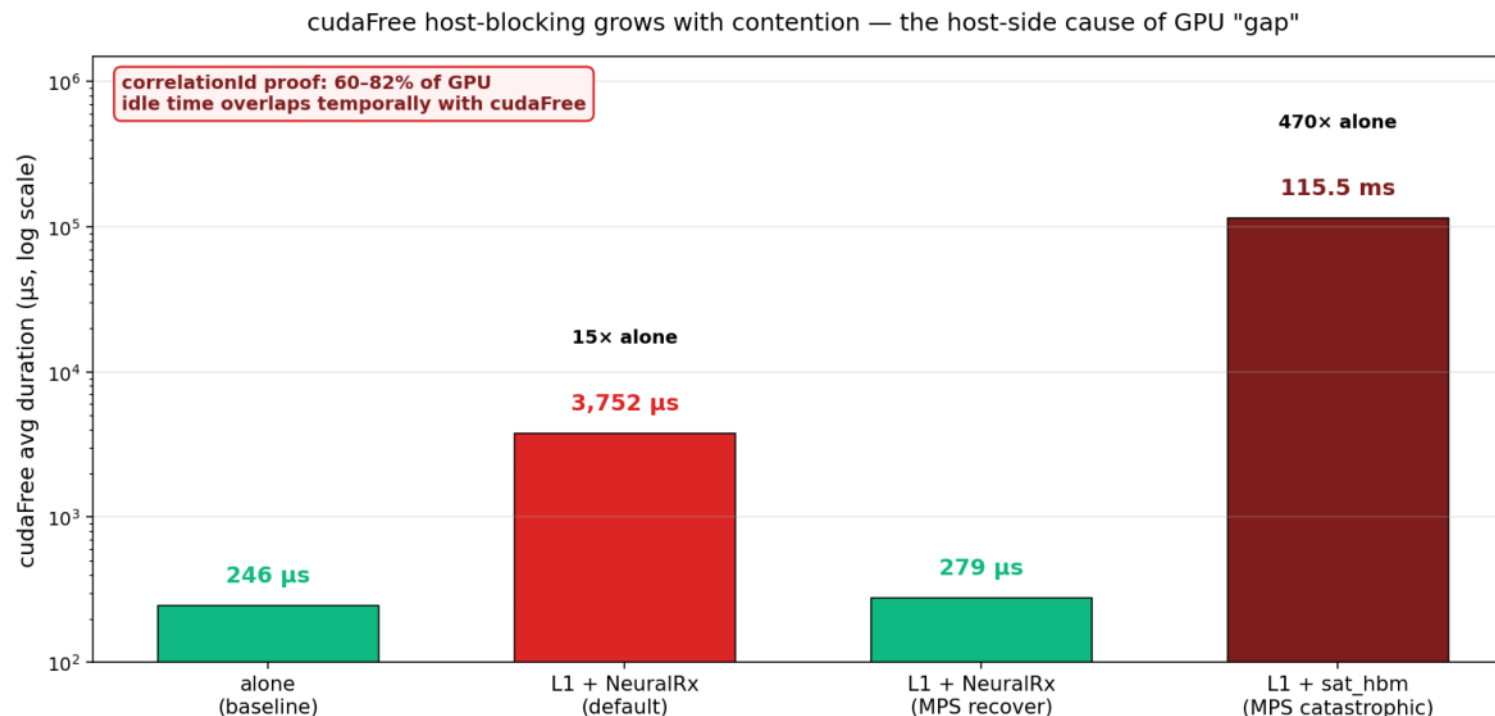
Mechanism Layer 2 — Op Count Stable, Duration Inflates

- **Call Count is Invariant:** memcpy and memset issue identical call counts (5,778 each) under alone and co-location; AI does not add memory work.
- **Per-Call Duration Grows by 133%:** memcpy average duration rises from 4.2 μ s to 9.8 μ s while the workload performs the same number of ops.
- **Rejects 'More Work' Hypothesis:** L1 is not asked to do more — it waits longer for the same work to complete, ruling out demand-driven explanations.



Mechanism Layer 3 — cudaFree Host-Blocking is the Direct Cause

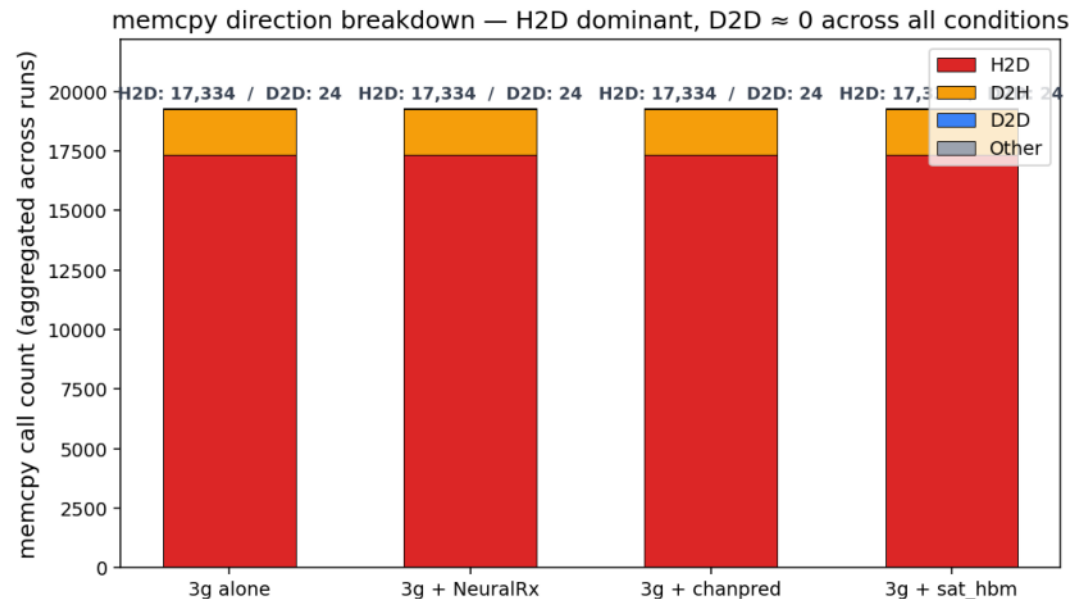
- **Host-Side cudaFree Inflates with Contention:** cudaFree average duration grows from 246 μ s (alone) to 3,752 μ s under NeuralRx (15 $\times$) and 115,506 μ s (115 ms) under MPS + sat_hbm (470 $\times$).
- **The Same Call Recovers Under MPS:** cudaFree returns to 279 μ s under MPS + NeuralRx — the host stops blocking only when concurrent execution frees the device.
- **Causal Proof via correlationId:** 60-82 % of GPU idle gap time overlaps temporally with the host's cudaFree call, identifying it as the direct cause of the GPU stall.



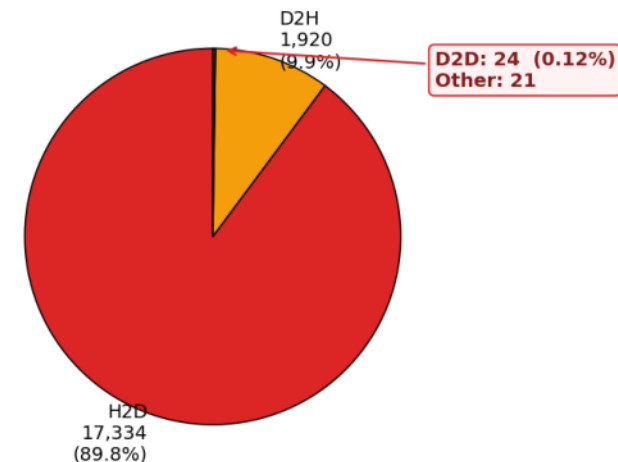
Mechanism Layer 4 — Chip-Wide PCIe/DMA Copy Engine

- **Direction Breakdown:** Across all measured memcpy calls, 89.8% are H2D, 9.9% are D2H, and only 0.12% are D2D.
- **Queue Location Identified:** With $D2D \approx 0$, the contention path is not at the HBM controller; it is at the chip-wide PCIe/DMA copy engine.
- **Single Shared Resource:** The PCIe/DMA copy engine is one physical unit serving the entire chip — partitioning fundamentally cannot isolate it.

Queue location proof — memcpy direction: 100% H2D, $D2D \approx 0$
→ Queue is at chip-wide PCIe/DMA copy engine (not partition-isolated HBM controller)

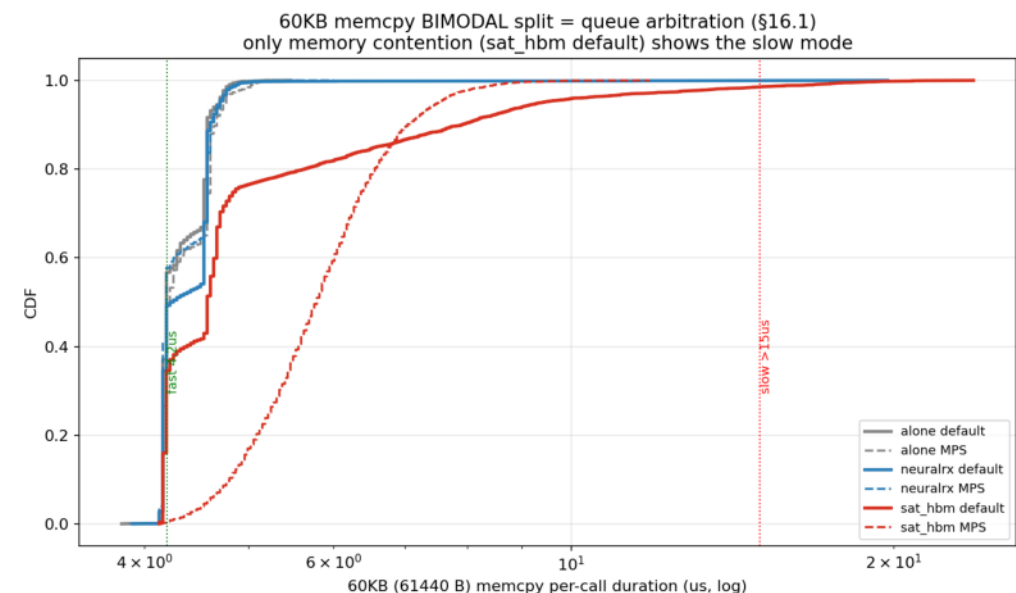
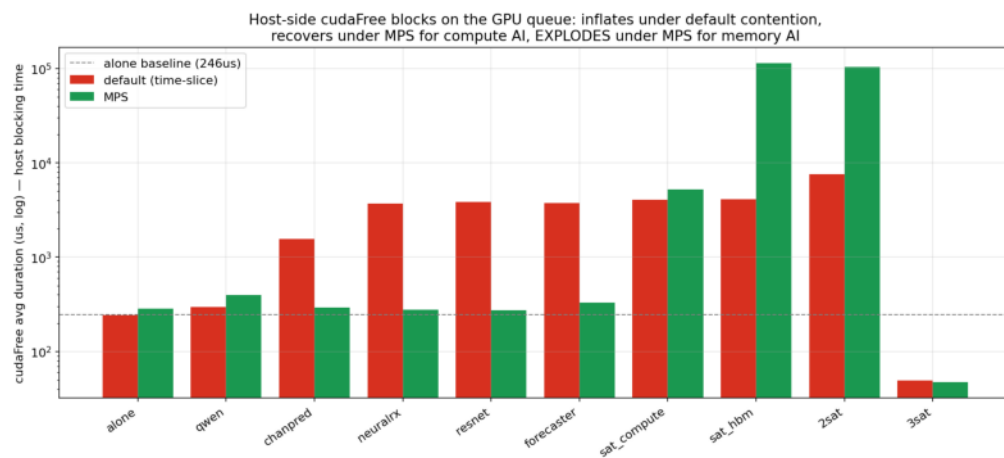


3g alone — proportional breakdown



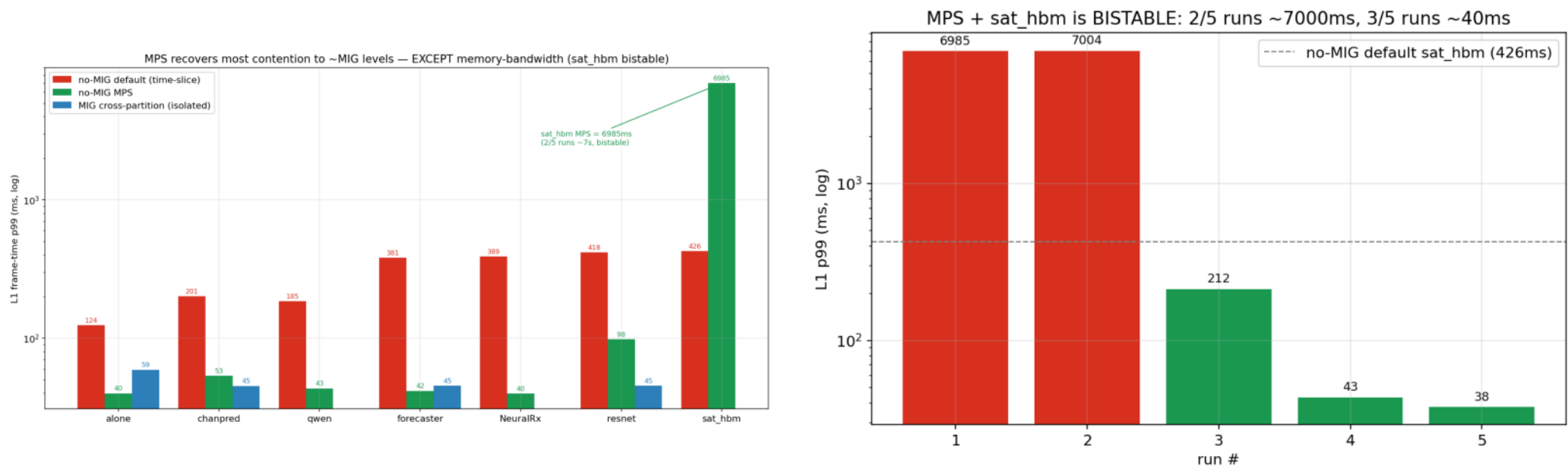
Cross-Platform Mechanism Validation

- **Bimodal Signature Reproduced:** Perlmutter no-MIG (different cluster, different driver) exhibits the same 60 KB memcpy bimodal split ($4.2 \leftrightarrow 16.8 \mu\text{s}$).
- **cudaFree Host-Blocking Scales with Contention:** cudaFree average grows from 246 μs (alone) to 3,752 μs under NeuralRx (15 $\times$) and 115,506 μs under MPS + sat_hbm (469 $\times$).
- **Causal Proof via correlationId:** 60-82% of GPU idle gap time aligns temporally with cudaFree blocking, establishing direct causality between host-side blocking and GPU stall.



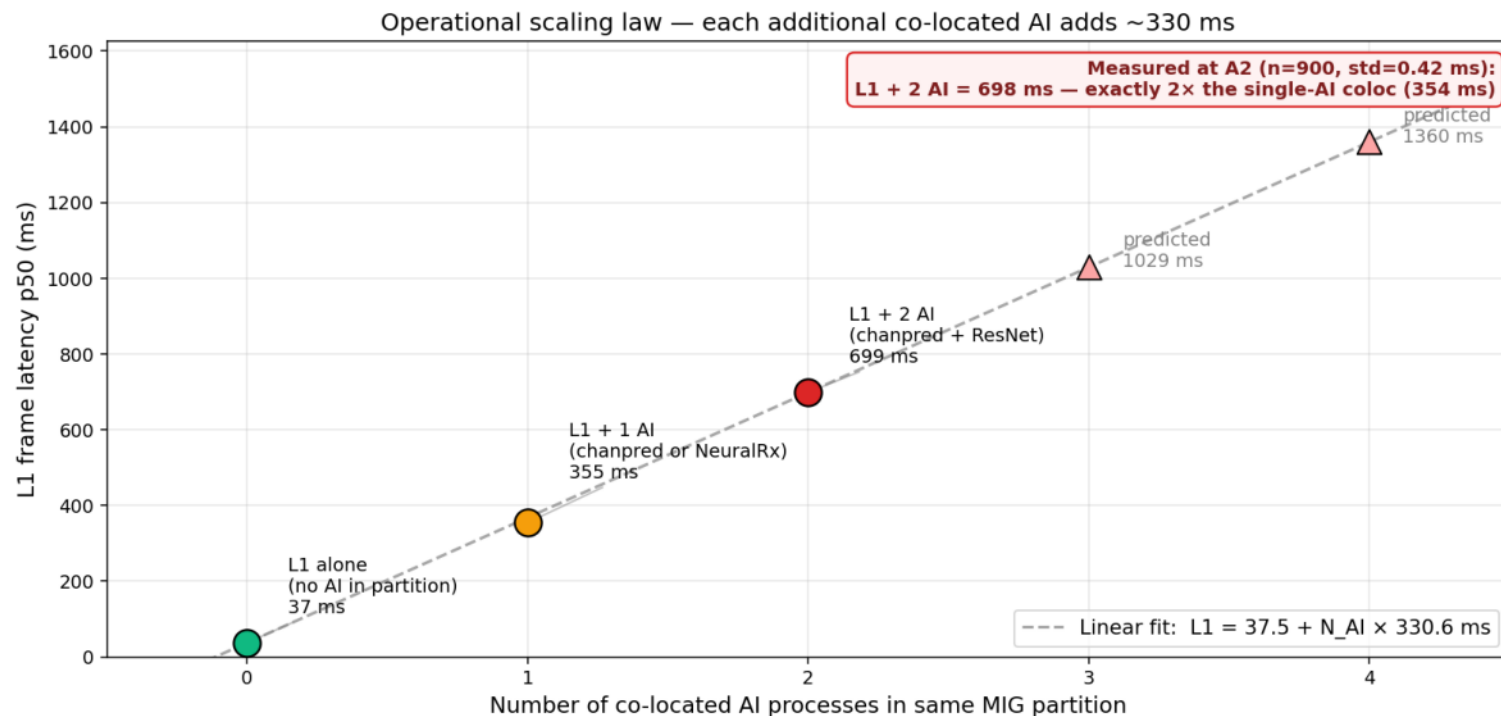
MPS Recovers Compute AI, Catastrophic for Memory AI

- **Compute-Bound AI Recovers Under MPS:** NeuralRx p99 drops from 389 ms to 40 ms — below MIG cross-partition (197 ms); forecaster (381→42 ms) and Qwen (185→43 ms) show the same pattern.
- **Memory-Bound AI Becomes Catastrophic:** sat_hbm under MPS rises from 426 ms to 6,985 ms (bistable across runs), worse than any other configuration measured.
- **Same Mechanism, Different Trigger:** MPS still routes through the same cudaFree host-blocking path; memory-bound workloads starve the device and re-engage the failure mode MPS was meant to bypass.



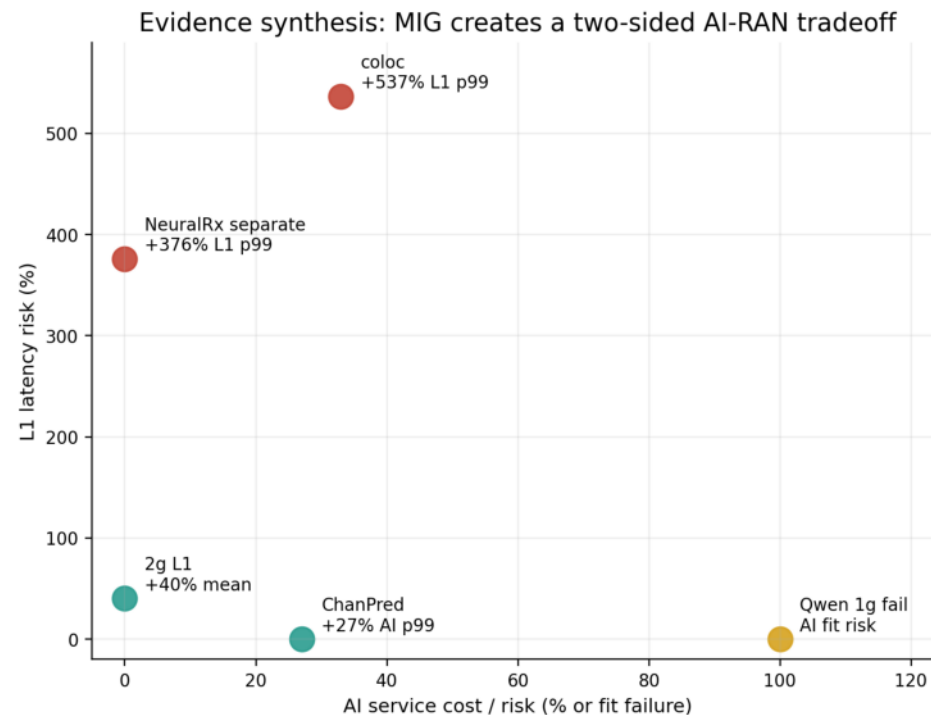
Operational Scaling Law — Each AI Adds ~330 ms

- **Linear Cost per Co-Located AI:** L1 frame latency follows $L1 = 37.5 + N_AI \times 330.6$ ms when AI processes share the same MIG partition — every additional tenant adds a fixed ~330 ms.
- **A2 Validates the Slope:** L1 + 2 AI measures 698 ms (n=900, std=0.42 ms) — exactly 2× the single-AI co-location (354 ms), confirming the linear model rather than a single-step collapse.
- **Multi-Tenant Misplacement Compounds:** Operational mistakes do not saturate; each additional misplaced tenant adds another fixed penalty, predicting 1,029 ms at 3 AI and 1,360 ms at 4 AI.



Cost Decomposition — Two Hardware-Level Costs

- **Structural Cost (memset):** Tied to partition HBM bandwidth share (2/7, 3/7, 4/7); selectable through partition sizing.
- **Contention Cost (memcpy):** Tied to chip-wide PCIe/DMA copy engine arbitration; not isolable by any partition choice.
- **Unified Across Three Platforms:** MIG, no-MIG, and MPS each avoid one cost while remaining exposed to the other — there is no single configuration that escapes both.



Design Rules & Next Steps

- **Placement Enforcement is Mandatory:** L1 and AI must reside in different MIG partitions; one misplacement produces the full 8× collapse.
- **Partition Sizing Serves Only L1 Budget:** Choose partition size for L1 SM/HBM headroom; sizing has no mitigating effect under co-location.
- **AI Workload Classification is Required:** Memory-bound AI cannot be safely co-located even under MPS, so the deployment policy must distinguish compute-bound from memory-bound tenants.
- **MIG $\neq$ Chip-Wide Isolation:** MIG isolates per-partition SM/HBM but not the chip-global PCIe/DMA copy engine — the dominant contention path remains outside its abstraction.

Thank You
